Alessio Olivieri

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Education

MSc in Artificial Intelligence & Robotics – Sapienza University of Rome Jan 2025 – May 2027 (expected)
Current average: 29/30. Coursework: Robotics, Machine Learning, AI, Deep Learning, NLP, Neural Networks, Optimization, HPC, Mobile & Medical Robotics.

BSc in Applied Computer Science & Artificial Intelligence – Sapienza Oct 2020 – Dec 2024
Final grade: 101/110. Coursework: Python, Java, R, Linear Algebra, Physics, Calculus I–II, Machine Learning, Computer Vision, Data Analysis, Cybersecurity, OS, Networks, Linear Programming, Algorithms, Probability, Statistics, Discrete Mathematics.

Research & Technical Projects

Vision Transformer Pruning via Mechanistic Interpretability code | slides
- Implemented ACDC-style causal analysis and activation patching in **PyTorch** to identify redundant components in a ViT.
- Pruned **85% of attention heads and 94% of the computation graph**; retrained the sparse architecture from scratch, achieving **2× faster CPU inference** with a **9% accuracy drop**.

NCCL Swing Collective Algorithm – Bachelor's Thesis code
- Implemented a GPU collective-communication algorithm using **NCCL / MSCCLang** and benchmarked multi-GPU execution, including on the **Leonardo supercomputer**.
- Measured speedups averaging **50%** and reaching **200%** in tested torus-network configurations.

Encrypted Network Traffic Classification – Research code
- Evaluated graph-based ML with **PyTorch Geometric** on ISCX-VPN: **31 captures, 1,674 eligible samples, 6 classes**; used flow-disjoint splits, randomized/header-only ablations, and **3 seeds**.
- Real-input baseline: **92.53% ± 1.51 accuracy**, macro-F1 **0.9168 ± 0.0165**; randomized: **73.60% ± 14.95**; header-only: **79.33% ± 8.62**. Audited capture provenance and documented why capture-independent six-class validation was unavailable.

Gadgetbridge / Samsung Gear S3 Android–Tizen Support code
- Independently built Bluetooth / Samsung Accessory Protocol integration for the Gear S3: phone notifications, replies, contacts, reconnect handling and action-ID stability.
- Debugged traffic and Android lifecycle failures on physical devices; implemented automated regression tests (**65 plus 12 additional tests** passed in the recorded validation).

Video Action Segmentation University team project
- Compared **MS-TCN, BiGRU and MS-CBiGRU** for frame-level action labeling; best **macro-F1 ≈ 0.83** and **Jaccard ≈ 0.71**.

Parallel Multi-Layer Perceptron (MLP) code
- Built an MLP from scratch in **C/Pthreads**; measured approximately **1,000% speedup with 10 CPU cores**. Implemented a **CUDA** version for parallel execution.

Forest Surveillance: Real-Time AI Fire Detection – 30L code | report
- Rewrote video processing in **NumPy**, improving throughput from **0.1 FPS to 60 FPS**; improved reported detection accuracy to **95%** using background subtraction.

WasaPhoto – Full-Stack Social Platform – 30L code
- Designed frontend, backend and database (**Go, Vue.js, SQLite, Docker**); built a **REST API with OpenAPI** for posts, follows and bans.

Ingredients / Recipe Network Analysis code | data
- Scraped **6,000+ recipes** from Giallo Zafferano; analyzed ingredient co-occurrence networks and identified frequently used ingredients.

AI Orders Chatbot / Natural Language to SQL code | demo
- Used prompt engineering with **Llama 8B / 70B** to turn user questions into **SQL queries**; deployed the interface with **Streamlit**.

Experience

Mathematics Tutor – Linear Algebra, Sapienza University of Rome Sep 2025 – Feb 2026
Delivered supplementary Linear Algebra classes alongside the main university course, supporting BSc Applied Computer Science & Artificial Intelligence (ACSAI) students at Sapienza.

Programming Tutor Nov 2023 – Jan 2024
Tutored engineering, AI, statistics and economics students; improved learning materials and relayed feedback to department heads.

Plum Juice – Technical Writer & Team Member Sep 2024 – Present
Wrote technical articles and managed social media (website).

Skills

Programming: Java, JavaScript, Python, Go, SQL, C, RISC-V, R; PyTorch, NumPy, CUDA, Pthreads, Docker, Git, Linux.
Problem solving: Scored **30L/30** in the Python exam featuring **12 challenging LeetCode-style problems**.
Languages: Italian (C2), English (C2), Spanish (A2), Polish (A2).